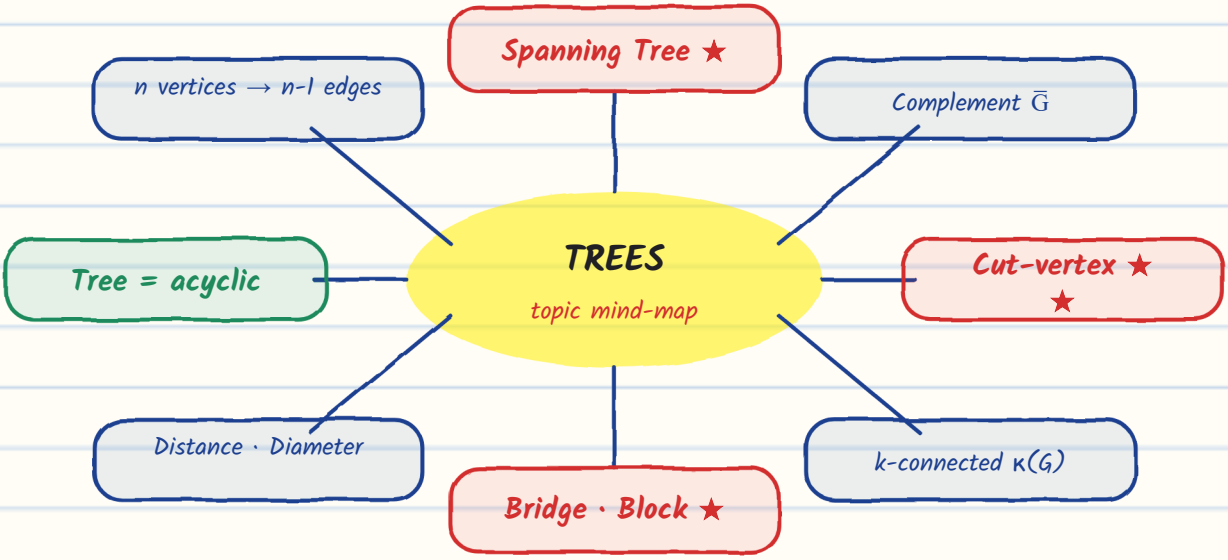
Trees & Graph Connectivity

Tree · Spanning tree · Distance · Complement · Cut-vertex · Bridge · k-connected: exam-ready in 10 sheets



↑ the whole topic on one page: red = high-yield, green = the one definition every question builds from

➜ Syllabus coverage

The syllabus line is "Graph theory: basic definitions — simple graph, directed graph, subgraph, degree, walk, path, cycle, special graphs (K_n, C_n, $K_{m,n}$), representations (adjacency matrix, adjacency list)". This lesson covers Trees, Spanning Trees, Distance, Complementary Graphs, Cut-Vertex, Bridges, k-Connected which build on the graph fundamentals.

High-yield list

① Tree: connected + acyclic
② n vertices → n−1 edges
③ Unique path between any two
④ Spanning tree = |V|−1 edges
⑤ dist(u,v) = shortest path
⑥ $G \cup \bar{G} = K_n$
⑦ Cut-vertex: removal ↑ comps
⑧ Bridge: not in any cycle
⑨ $\kappa \le \lambda \le \delta$ (Whitney)
⑩ $\kappa(K_n) = n-1$

Legend yellow = key term · red circle = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

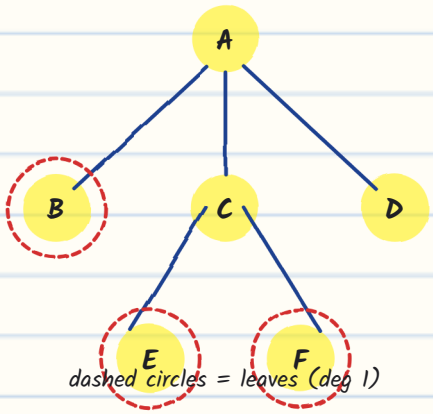
➜ Contents

✓ Tree definition & properties … p. 2
✓ Spanning trees & Cayley's formula … p. 3
✓ Distance, eccentricity, diameter … p. 4
✓ Complementary graphs … p. 5
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1 Trees

definition ★

Definition A tree is a **connected, acyclic** undirected graph. Equivalently: a graph with no cycles in which every pair of vertices is joined by exactly one path. A forest is a disjoint union of trees — an acyclic graph whose components are trees.



A tree with 6 vertices and 5 edges; B, E, F are leaves

Property	Description
Edges	If a tree has n vertices, it has $n - 1$ edges
Unique path	Exactly one path between any two vertices
Adding edge	Adding any edge creates exactly one cycle
Removing edge	Removing any edge disconnects the graph
Leaves	Every tree with $n \geq 2$ has at least 2 leaves (deg 1)
Forest	n vertices, k trees → $n - k$ edges

Memory trick "**A tree never loops — it only branches.**" No cycles, no shortcuts, just one road between any two nodes. That is why trees are the skeleton of every network:
~~minimum edges for maximum reach.~~

Exam trap (i) A tree is **not** just "acyclic" — it must also be **connected**. An acyclic graph with 4 vertices and 2 edges is a forest, not a tree. (ii) A single vertex ($n = 1$) is a tree with 0 edges — the formula $n - 1$ still holds. (iii) "Leaves = degree 1" — the root of a rooted tree may have degree 1 and still count as a leaf in the underlying unrooted tree.

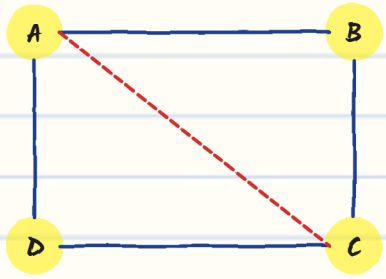
Worked example ✍ verify tree properties A graph has 8 vertices and 7 edges. Is it necessarily a tree? **No** — it could be disconnected (e.g. a triangle + 5 isolated vertices has 3 edges, not 7). A connected graph with n vertices and $n - 1$ edges **is** a tree. Here: if we confirm connectivity, then yes — 8 vertices, 7 edges, connected ⟹ tree.

2 Spanning Trees application ★

Definition A spanning tree of a connected graph $G = (V, E)$ is a subgraph $T = (V, E')$ such that: (i) T includes **all vertices** of G, and (ii) T is a tree (connected and acyclic). Any spanning tree has exactly $|V| - 1$ edges.

Key facts

- ✓ Every connected graph has at least one spanning tree
- ✓ A graph with cycles can have **many** spanning trees
- ✓ Number of edges in any spanning tree = $|V| - 1$
- ✓ Removing one edge from each cycle in G yields a spanning tree



dashed = one possible removal

C_4: removing any one of 4 edges gives a spanning tree; 4 total

Cayley's formula The number of distinct **labelled** spanning trees of the complete graph K_n is n^{n-2}. For K_4 this gives $4^2 = 16$ spanning trees. For K_5 this gives $5^3 = 125$.

Worked example ✍ spanning trees of C_4

A cycle C_4 has 4 vertices and 4 edges. Removing any one edge breaks the cycle while keeping the graph connected, giving a spanning tree. Since there are 4 edges, C_4 has exactly **4** spanning trees. For K_4 (6 edges), Cayley gives $4^2 = 16$ labelled spanning trees.

Exam trap (i) A spanning tree of G is **not** a subgraph of every spanning tree — it is a spanning tree **of** G. Different spanning trees share vertices but may differ in edges. (ii) A disconnected graph has **no** spanning tree — only a spanning forest. (iii) K_4 has $4^{4-2} = 16$ spanning trees, not 6. K_5 has $5^3 = 125$.

Memory trick "**Spanning tree = skeleton.**" It touches every vertex (spans all) and uses the fewest possible edges (tree) to keep everything connected. Think of it as the minimum-cable wiring to link all computers in a network.

IT link Minimum-spanning-tree algorithms (Kruskal's, Prim's) are used in network design, cluster analysis, and approximating the Traveling Salesman Problem.

3 Distance in Graphs

metrics ★

Definition The **distance** $d(u, v)$ between two vertices u and v is the length (number of edges) of a **shortest path** connecting them. If no path exists, $d(u, v) = \infty$.

Property	Description
$d(u, v) = 0$	If and only if $u = v$
$d(u, v) = d(v, u)$	Symmetric in undirected graphs
Triangle inequality	$d(u, w) \leq d(u, v) + d(v, w)$
Infinite distance	If u and v are in different components

Derived metrics

- ✓ **Eccentricity:** $ecc(v) = \max\{d(v, u) \mid u \in V\}$
- ✓ **Diameter:** $diam(G) = \max\{ecc(v) \mid v \in V\}$
- ✓ **Radius:** $rad(G) = \min\{ecc(v) \mid v \in V\}$
- ✓ **Center:** $\{v \mid ecc(v) = rad(G)\}$
- ✓ **Periphery:** $\{v \mid ecc(v) = diam(G)\}$



green circle = center vertex C

P_5: $ecc(A)=4$, $ecc(B)=3$, $ecc(C)=2$, $ecc(D)=3$, $ecc(E)=4$; $diam=4$, $rad=2$

Worked example ✍ P_5 path graph

Vertices A–B–C–D–E in a line. Distances from A: $d(A,B)=1$, $d(A,C)=2$, $d(A,D)=3$, $d(A,E)=4$, so $ecc(A)=4$. Similarly $ecc(B)=3$, $ecc(C)=2$, $ecc(D)=3$, $ecc(E)=4$. Diameter = max ecc = 4. Radius = min ecc = 2. Center = $\{C\}$. Periphery = $\{A, E\}$.

Exam trap (i) Diameter is the **maximum** eccentricity, not the maximum degree. (ii) The center may contain **one or two adjacent** vertices — e.g. P_4 has center $\{B, C\}$. (iii) In a disconnected graph, distance is ∞ between components, so diameter is often defined only for connected graphs.

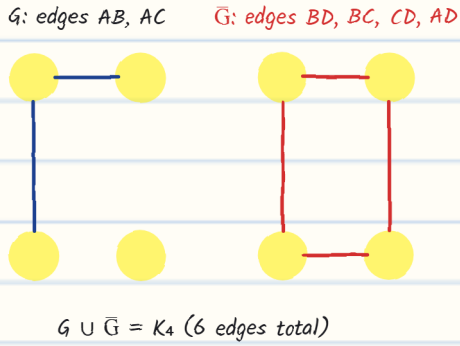
Worked example ✍ star graph S_4 Center vertex c joined to leaves a, b, d. Distances: $d(c,a)=d(c,b)=d(c,d)=1$, $d(a,b)=d(a,d)=d(b,d)=2$. Eccentricities: $ecc(c)=1$, $ecc(a)=ecc(b)=ecc(d)=2$. Diameter = 2, Radius = 1, Center = $\{c\}$, Periphery = $\{a, b, d\}$. Note: radius ≠ diameter here — only paths and cycles can have rad = diam/2 approximately.

4 Complementary Graphs

construction ★

Definition The **complement** of a graph $G = (V, E)$, denoted $\overline{G}$, has the **same vertex set** V, and an edge $\{u, v\}$ is in $\overline{G}$ **if and only if** $\{u, v\}$ is **not** in E. Thus G and $\overline{G}$ partition the edges of the complete graph K_n.

Property	Description
$G \cup \overline{G}$	Complete graph K_n (edge-disjoint union)
$\overline{\overline{G}}$	G (complement of complement is original)
Edges	$\lvert E(\overline{G})\rvert = n(n-1)/2 - \lvert E(G)\rvert$
Self-compl.	$G \cong \overline{G}$; requires $n \equiv 0$ or $1 \pmod 4$



$G \cup \overline{G} = K_4$ (6 edges total)

G has 2 edges, $\overline{G}$ has 4; together they form K_4 with 6 edges = 4·3/2

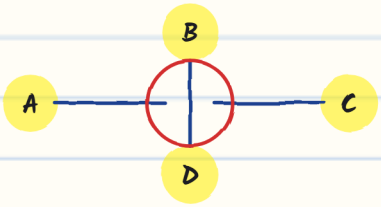
Self-complementary A graph isomorphic to its complement. Example: P_4 (path on 4 vertices) is self-complementary. For a self-complementary graph on n vertices, $|E| = n(n-1)/4$. A self-complementary graph on 8 vertices has $8 \cdot 7/4 = 14$ edges.

Exam trap (i) G and $\overline{G}$ **never** share an edge — their edge sets are complementary within K_n. (ii) Both G and $\overline{G}$ cannot be disconnected for $n \geq 2$ — at least one is connected. (iii) A self-complementary graph on n vertices has exactly $n(n-1)/4$ edges, so $n(n-1)$ must be divisible by 4 — this means $n \equiv 0$ or $1 \pmod 4$.

Memory trick "**Complement = the rest.**" Take K_n, remove G's edges, what remains is $\overline{G}$. For a self-complementary graph, G and $\overline{G}$ are mirror images in edge space — each gets exactly half of K_n's edges.

S Cut-Vertices (Articulation Points) weak point ★

Definition A vertex v is a **cut-vertex** (articulation point) if removing v and all incident edges **increases** the number of connected components. A graph with no cut-vertices is **2-connected** (or non-separable).



B is a cut-vertex (red circle)

Removing B leaves {A}, {C}, {D} — 3 components; B is a cut-vertex

Key properties

- ✓ Bridges and cut-vertices are **weak points** in networks
- ✓ Leaves are **not** cut-vertices (except in K_2)
- ✓ Every cycle has **no** cut-vertices
- ✓ A block is a maximal subgraph **without** cut-vertices
- ✓ A tree with $n \geq 3$ has $n - 2$ cut-vertices (all non-leaf vertices)

Exam trap A leaf (degree-1 vertex) is **never** a cut-vertex in a tree with ≥ 3 vertices — removing it leaves the rest still connected. Only in K_2 does the single edge's endpoint count as a cut-vertex. Also: a graph with a cut-vertex has $\kappa(G) = 1$, so it is **not** 2-connected.

Memory trick "**A cut-vertex is a bottleneck.**" Remove it and the network falls apart into more pieces. Think of a bridge city on a river — close it and the two banks are isolated.

Worked example ✍ find cut-vertices

Graph: A—B—C with D attached to B (B is centre). Removing B gives components {A}, {C}, {D} — **3 components** (was 1). So B is a cut-vertex. A, C, D are leaves — removing any one leaves 2 vertices still connected, so they are **not** cut-vertices. In a path P_5 (A–B–C–D–E), the cut-vertices are B, C, D (3 of them); A and E are leaves, not cut-vertices.

IT link Cut-vertices identify single points of failure in communication networks, power grids, and road systems — locations where one outage isolates entire regions.

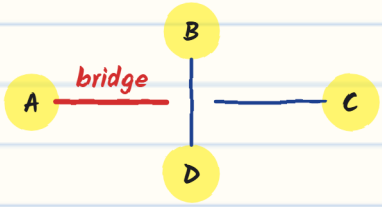
6 Bridges (Cut-Edges)

critical edge ★

Definition An edge e is a bridge (cut-edge) if removing e **increases** the number of connected components. Equivalently: e is a bridge **if and only if** it belongs to **no cycle** in G.

Key facts

- ✓ Every edge in a tree is a bridge
- ✓ An edge is a bridge ⟺ it is not in any cycle
- ✓ Removing a bridge disconnects the graph
- ✓ Bridges are the edge-analogue of cut-vertices
- ✓ A graph with no bridges is **2-edge-connected**



every edge of a tree is a bridge

A–B is a bridge – like every edge of this tree

DFS bridge-finding algorithm Compute **discovery time** disc[u] and **low-link** value low[u] = min(discover time of u, discover time of ancestor reachable via back-edge from subtree of u). Edge (u, v) (u = parent of v) is a bridge if low[v] > disc[u].

Exam trap (i) An edge in a **cycle** is **never** a bridge — removing one edge from a cycle leaves a path, still connected. (ii) Every tree edge is a bridge, but not every bridge is in a tree — bridges exist in graphs with cycles too. (iii) low[v] > disc[u] (strictly greater); low[v] = disc[u] means there is a back-edge, so (u,v) is **not** a bridge.

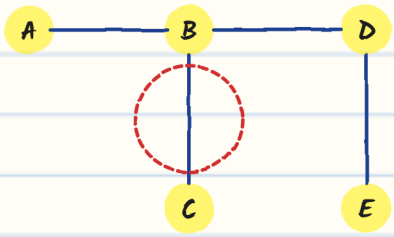
Memory trick "**No cycle = bridge.**" If you can绕 (go around) from v back to u without using edge (u,v), then (u,v) is safe — it is not a bridge. If not, it is a bridge.

Worked example ✍ bridges in a triangle with a tail Graph: A—B—C forms a triangle (A-B, B-C, C-A), and C—D is a tail. Edge C—D is a bridge (removing it isolates D). Edges A-B, B-C, C-A are **not** bridges — each lies on the triangle cycle. Removing any one triangle edge leaves a path A-B-C, still connected. So this graph has exactly 1 bridge : C—D.

7 Blocks & k-Connected Graphs

connectivity ★

Block A block is a maximal connected subgraph with **no cut-vertex**. Blocks are either: (i) a maximal 2-connected subgraph, (ii) a bridge (single edge that is a block), or (iii) an isolated vertex. A block tree (BC-tree) has nodes for blocks and cut-vertices, with edges linking each cut-vertex to the blocks it belongs to.



B is a cut-vertex; blocks: {A,B}, {B,C,D}, {D,E}

Block decomposition: three blocks joined at cut-vertices B and D

k-connected A graph is k-connected if $|V| > k$ and removing any set of fewer than k vertices leaves the graph connected. The vertex connectivity $\kappa(G)$ is the minimum number of vertices whose removal disconnects G (or makes it trivial).

Edge connectivity The edge connectivity $\lambda(G)$ is the minimum number of edges whose removal disconnects G. **Whitney's Theorem:** $\kappa(G) \le \lambda(G) \le \delta(G)$, where $\delta(G)$ is the minimum degree.

Exam trap (i) $\kappa(K_n) = n - 1$, **not** n. You need to remove all but one vertex to disconnect K_n. (ii) A graph with a cut-vertex has $\kappa = 1$ — it is **not** 2-connected. (iii) $\kappa \le \lambda \le \delta$ is always true; equality on both sides holds for K_n and C_n but not generally.

Memory trick "**κ is the weakest link, δ is the strongest ceiling.**" Vertex connectivity (κ) ≤ edge connectivity (λ) ≤ minimum degree (δ). The minimum degree is an upper bound — you can always disconnect by removing all neighbours of a min-degree vertex, but sometimes fewer suffice.

Worked example ✍ connectivity values

(a) Tree with $n \ge 2$: $\kappa = 1$ (has cut-vertices), $\lambda = 1$ (has bridges). (b) Cycle C_n, $n \ge 3$: $\kappa = 2$, $\lambda = 2$ — removing any 1 vertex or edge leaves a path (connected). (c) K_6: $\kappa = 5$, $\lambda = 5$ — you must remove 5 vertices/edges to isolate one vertex. (d) $K_{3,3}$: $\kappa = 3$, $\lambda = 3$ (3-regular, each vertex has degree 3).

Night-Before-the-Exam Sheet

Fact sheet

Tree: connected + acyclic	n vertices → $n-1$ edges
Unique path between any 2 vertices	Adding edge → exactly one cycle
Removing edge → disconnects	Leaves: at least 2 when $n \geq 2$
Forest: n vertices, k trees → $n-k$ edges	Spanning tree: $\|V\|-1$ edges
Cayley: K_n has n^{n-2} spanning trees	Distance: shortest-path length
Diameter = max eccentricity	Radius = min eccentricity
Center: vertices with ecc = rad	Periphery: vertices with ecc = diam
$G \cup \bar{G} = K_n$ (edge-disjoint)	$\|E(\bar{G})\| = n(n-1)/2 - \|E(G)\|$
Self-complement: $n \equiv 0,1 \pmod 4$	Cut-vertex: removal ↑ components
Bridge: edge not in any cycle	Bridge test: low[v] > disc[u]
Block: maximal subgraph, no cut-v	$\kappa(G) \leq \lambda(G) \leq \delta(G)$ (Whitney)
$\kappa(K_n) = n-1$; $\kappa(C_n) = 2$	k-connected: survives removal of $k-1$ verts

→ Top 12 one-liners for MCQs

1. Tree: n vertices, $n-1$ edges
2. Forest: n vertices, k trees → $n-k$ edges
3. Spanning tree of G has $\|V\|-1$ edges
4. Cayley: K_n has n^{n-2} spanning trees
5. $d(u,v)$ = length of shortest path
6. diam(G) = max ecc; rad(G) = min ecc
7. $\|E(\bar{G})\| = n(n-1)/2 - \|E(G)\|$
8. Cut-vertex: removal increases components
9. Bridge: edge not in any cycle
10. Bridge test: low[v] > disc[u]
11. Block: maximal subgraph with no cut-v
12. Whitney: $\kappa \leq \lambda \leq \delta$

Top traps "Tree" requires both connected AND acyclic — acyclic alone is a forest · A leaf is never a cut-vertex in $n \geq 3$ · low[v] > disc[u] is strict — equality means a back-edge exists · Self-complementary needs $n(n-1)/4$ edges, so $n \equiv 0$ or $1 \pmod 4$ · $\kappa(K_6) = 5$, not 6 · Diameter is max eccentricity, not max degree · Center of P_4 is $\{B,C\}$, not a single vertex

All tricks together "A tree never loops — it only branches" · "Spanning tree = skeleton" · "Complement = the rest" · "A cut-vertex is a bottleneck" · "No cycle = bridge" · "κ is the weakest link, δ is the strongest ceiling".

✓ Practice MCQs (exam pattern, 10 marks)

try first, then check

1. How many edges does a tree with 20 vertices have?

(a) 20 (b) 21
(c) 18 (d) 19

2. Which is a bridge of a graph?

(a) An edge joining a vertex to itself (b) An edge between max-degree vertices
(c) An edge on every cycle (d) An edge whose removal ↑ components

3. G has 8 vertices and 10 edges. How many edges does $\bar{G}$ have?

(a) 10 (b) 28
(c) 38 (d) 18

4. A forest has 15 vertices and 4 trees. How many edges?

(a) 14 (b) 11
(c) 15 (d) 19

5. How many distinct labeled spanning trees does the complete graph K_4 have?

(a) 4 (b) 12
(c) 16 (d) 64

6. Which is true for every tree with ≥ 2 vertices?

(a) It contains a cycle (b) Every vertex is a cut-vertex
(c) It has exactly n edges (d) Every edge is a bridge

7. What is $\kappa(K_6)$, the vertex connectivity of K_6?

(a) 6 (b) 5
(c) 4 (d) 15

8. For path graph P_7, what are the radius and diameter?

(a) 3 and 6 (b) 6 and 3
(c) 3.5 and 7 (d) 4 and 6

9. For any graph G, which inequality (Whitney's) always holds?

(a) $\lambda \le \kappa \le \delta$ (b) $\delta \le \kappa \le \lambda$
(c) $\kappa \le \lambda \le \delta$ (d) $\kappa \le \delta \le \lambda$

10. A self-complementary graph with 8 vertices has how many edges?

(a) 14 (b) 28
(c) 7 (d) 16

Answer key → 1-d · 2-d · 3-d · 4-b · 5-c · 6-d · 7-b · 8-a · 9-c · 10-a

Written question Prove that every tree with n vertices has exactly $n - 1$ edges. 10

1. **Base case ($n = 1$).** A tree with one vertex has 0 edges. Formula: $1 - 1 = 0$. ✓ 2
2. **Inductive hypothesis.** Assume every tree with k vertices has $k - 1$ edges.
3. **Inductive step.** Let T be a tree with $k + 1$ vertices. T has at least two leaves (vertices of degree 1). Remove a leaf v and its incident edge e. The remaining graph T' has k vertices and is still a tree (removing a leaf preserves connectivity and acyclicity). By the hypothesis, T' has $k - 1$ edges. Therefore T has $(k - 1) + 1 = k = (k + 1) - 1$ edges. ✓ 5
4. **Conclusion.** By mathematical induction, every tree with n vertices has exactly $n - 1$ edges for all $n \ge 1$. 2